

# K. SLOAN LOUDEN-ROBINS

## ELECTRICAL ENGINEER

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### EDUCATION

December 2025

Texas State University – Ingram School of Engineering | ABET Accredited Program | San Marcos, TX

B.S. Electrical Engineering (Computer Engineering Concentration)

B.S. Applied Mathematics

Minor in Computer Science

### RELEVANT COURSEWORK

Circuits I & II, Electronics I & II, Machine Learning, VLSI Design, C++ I & II, Microprocessors, Linear Control Systems, Communication Networks, Power Electronics

### TECHNICAL SKILLS AND CERTIFICATIONS

- Engineering - LT Spice, Microwind, MATLAB, KiCad, Anaconda
- Languages - C++; Python; Unix/BASH
- Software – KiCad, Microsoft Office Suite, Visio
- Certifications - FE scheduled for November 2026

### PROJECTS

#### Project Manager / Data Processing Lead

January 2025 – December 2025

Foundation Monitoring System, Ingram School of Engineering

- Industry Choice Award at TXST Design Day
- Led project schedules, task delegation, sponsor coordination, documentation, and BOM/cost tracking.
- Developed Python data validation, filtering, and sensor analysis tools.
- Evaluated filtering, sensor scoring, and machine-learning methods.
- Created data-processing SOP for project handoff.

#### Co-researcher

September 2025 – December 2025

Scatter Detector, TXST Space Lab

- Researched scatter detection and nephelometry for a high-altitude balloon payload.
- Designed PCB in KiCad and scatter chamber in SolidWorks.
- Developed firmware for CO<sub>2</sub> measurement and data collection.
- Developed project timeline and technical proposal.

#### Lead Developer

February 2025 – April 2025

PID, Ingram School of Engineering

- Designed and simulated an analog PID controller for a 3rd-order system in MATLAB and LTspice.
- Optimized closed-loop performance using root locus and PID tuning techniques.
- Achieved 58.9% faster response, 39.7% lower settling time, and <2.1% overshoot.

### EXPERIENCE

#### Undergraduate Research Assistant

January 2024 – May 2024

Mathematics Department Texas State University, San Marcos, TX

- Researched sparse matrices and Fourier series decomposition with faculty advisor.

#### Lead Stylist

April 2013 – Present

LoLa Beauty, Austin, TX

- Received Glammy's award in 2017 and Best of the Knot Hall of Fame 2015-2024.
- Coordinated stylists, vendors, clients, and event schedules.

#### Learning Ambassador

June 2020 – June 2021

Amazon Warehouse (DAU2), Austin, TX

- Collected workflow data and employee feedback to identify process improvements.
- Trained and directed ~100 employees supporting 60K+ daily packages.